



Intracellular accumulations

Lec. 3

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Intracellular accumulations

Under some circumstances, cells may accumulate abnormal amounts of various substances, which may be harmless or may cause varying degrees of injury.

The substance may be located in the :

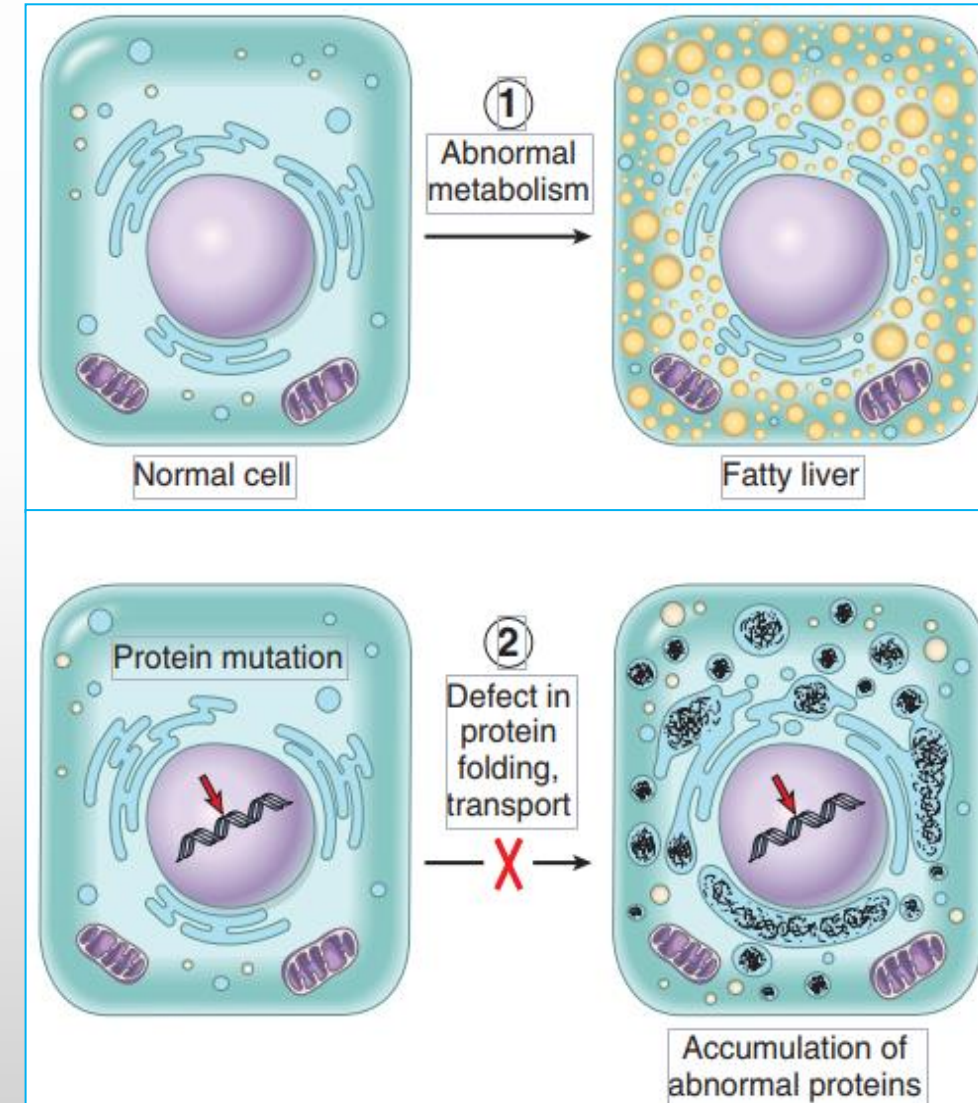
- 1) Cytoplasm
- 2) within organelles (typically lysosomes)
- 3) in the nucleus

The substance came to the cell through :

- 1) synthesized by the affected cells
- 2) produced elsewhere

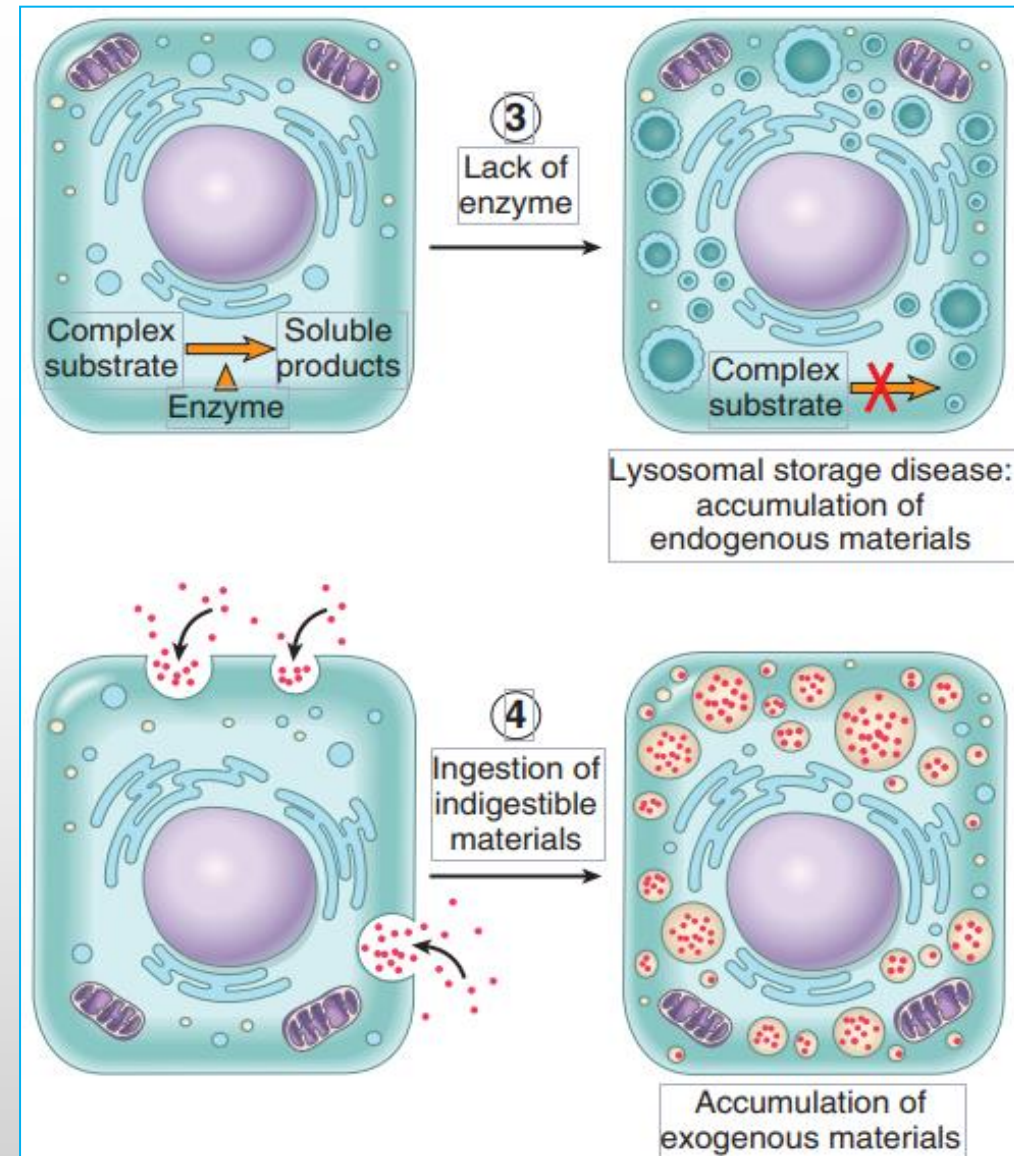
Mechanisms of intracellular accumulation

- (1) Abnormal metabolism, as in fatty change in the liver.
- (2) Mutations causing alterations in protein folding and transport, so that defective molecules accumulate intracellularly.

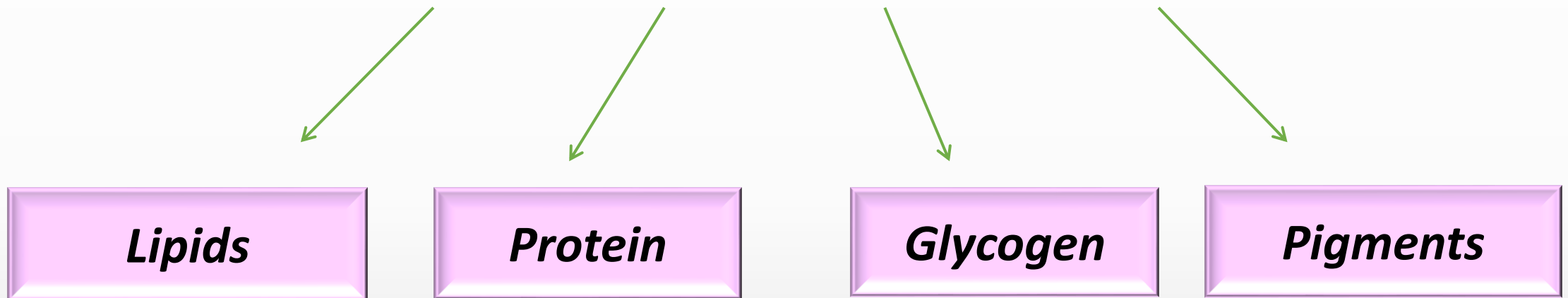


- (3) A deficiency of critical enzymes responsible for breaking down certain compounds, causing substrates to accumulate in lysosomes, as in lysosomal storage diseases.

- (4) An inability to degrade phagocytosed particles, as in carbon pigment accumulation.



Examples of intracellular accumulations



Triglycerides
Cholesterol and
cholesterol ester

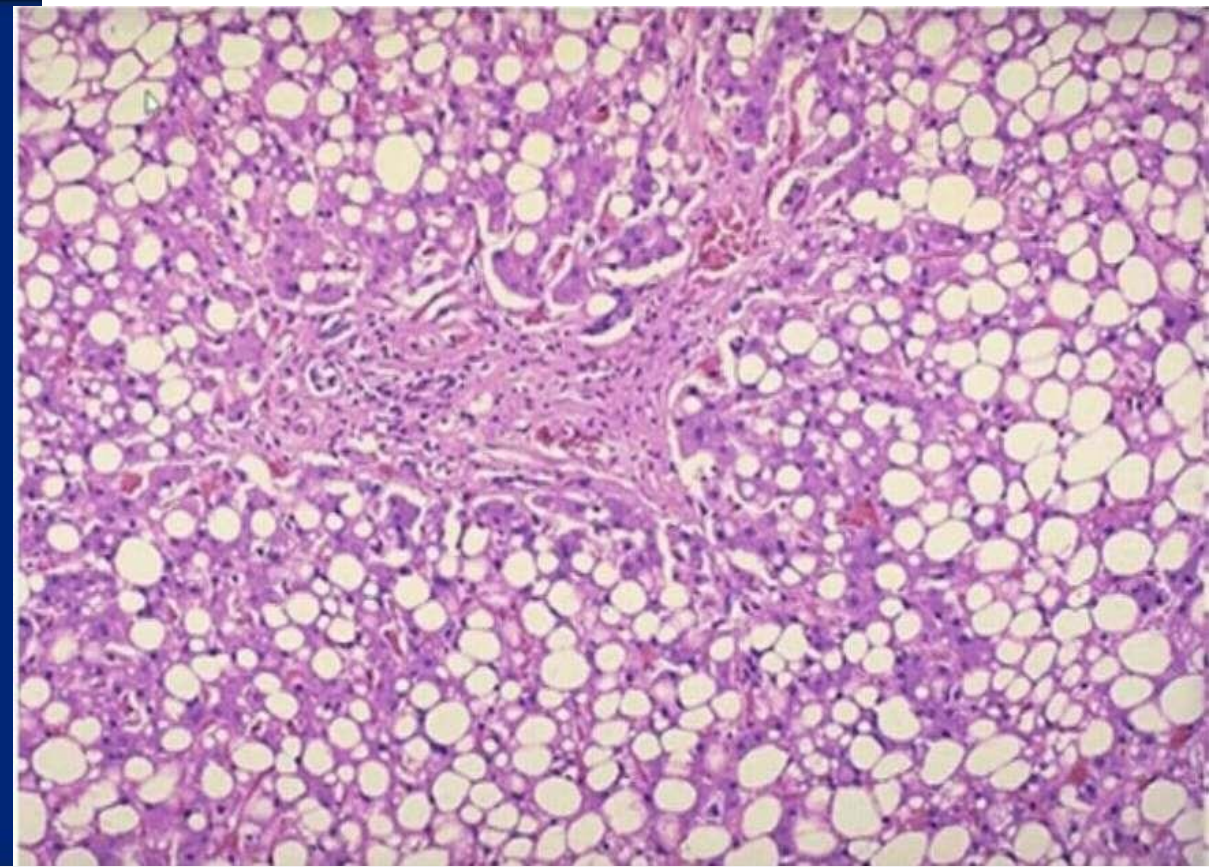
Fatty Change

- Fatty change, also called steatosis, refers to any accumulation of triglycerides within parenchymal cells. It is most often seen in the liver, since this is the major organ involved in fat metabolism, but also may occur in heart, skeletal muscle, kidney, and other organs.
- **Causes of fatty change :**
 - 1) Toxins
 - 2) protein malnutrition
 - 3) diabetes mellitus
 - 4) obesity
 - 5) anoxia



Sever fatty change liver

Gross features : the organ enlarges and become increasingly yellow, soft and greasy to touch .

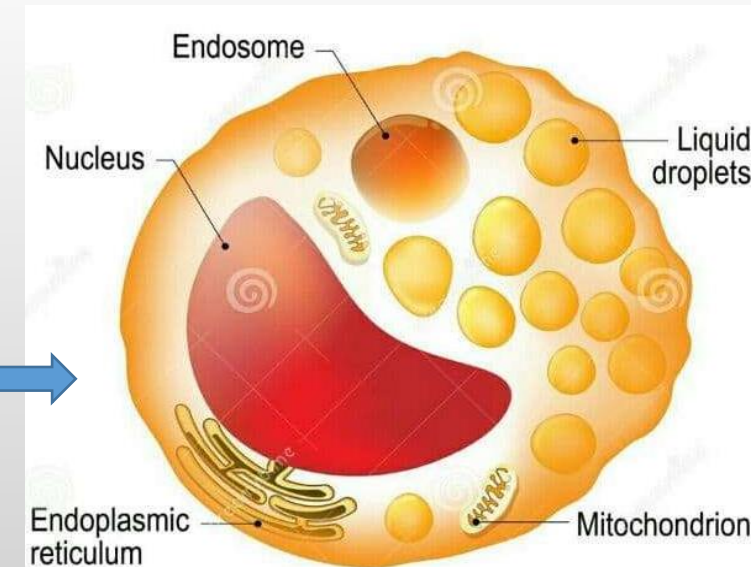
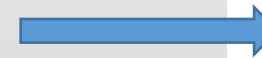


A significant number of hepatocytes show **fatty change** . The accumulated fat is represented by a large clear vacuole that pushes the nucleus to the extreme periphery of the cell .

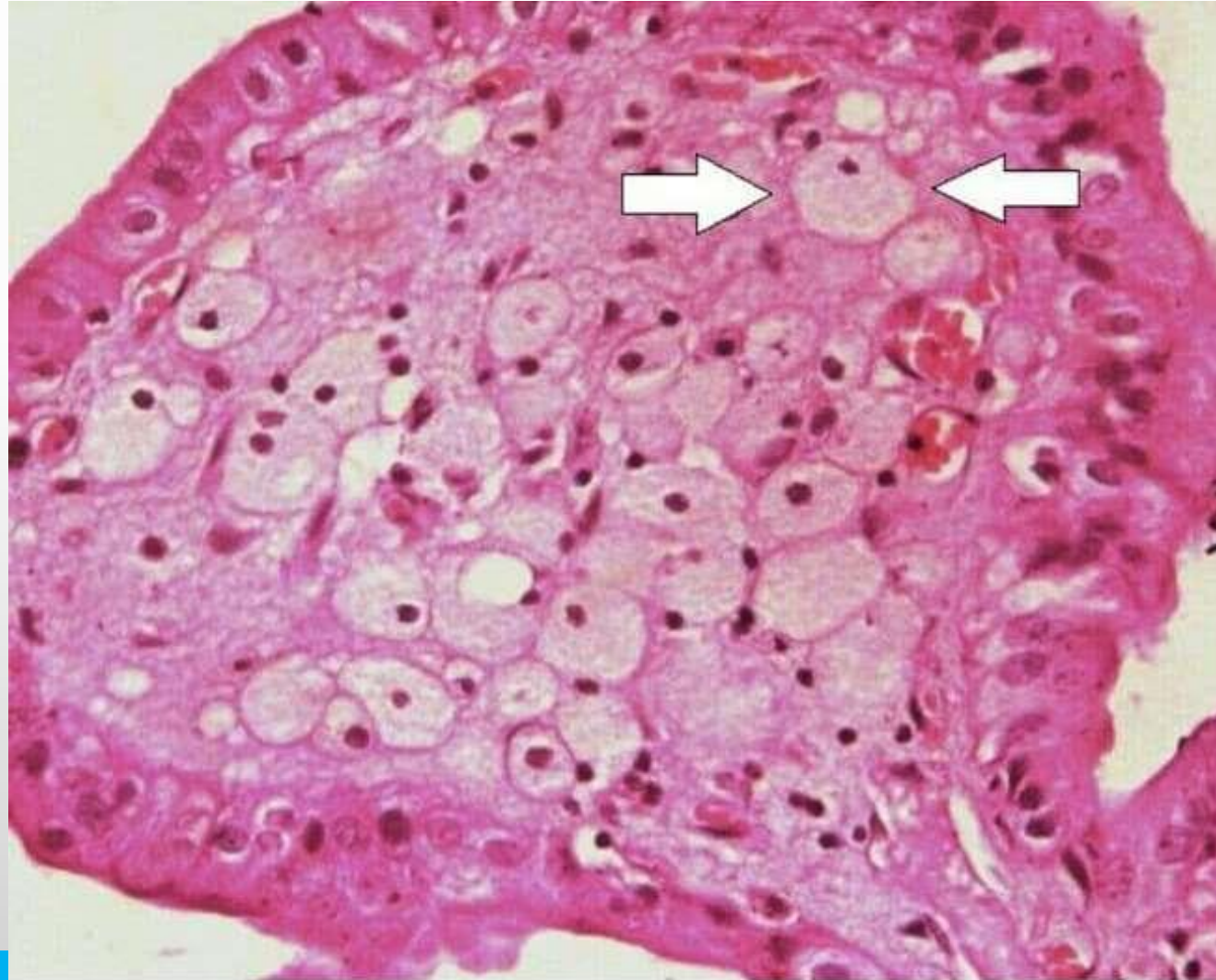
Cholesterol and Cholesterol Esters

- Phagocytic cells (macrophages) may become overloaded with lipid (triglycerides, cholesterol, and cholesteryl esters) in several different pathologic processes, of these, atherosclerosis is the most important . These macrophages may be filled with minute vacuoles of lipid, giving a foamy appearance to their cytoplasm called **foam cells**.

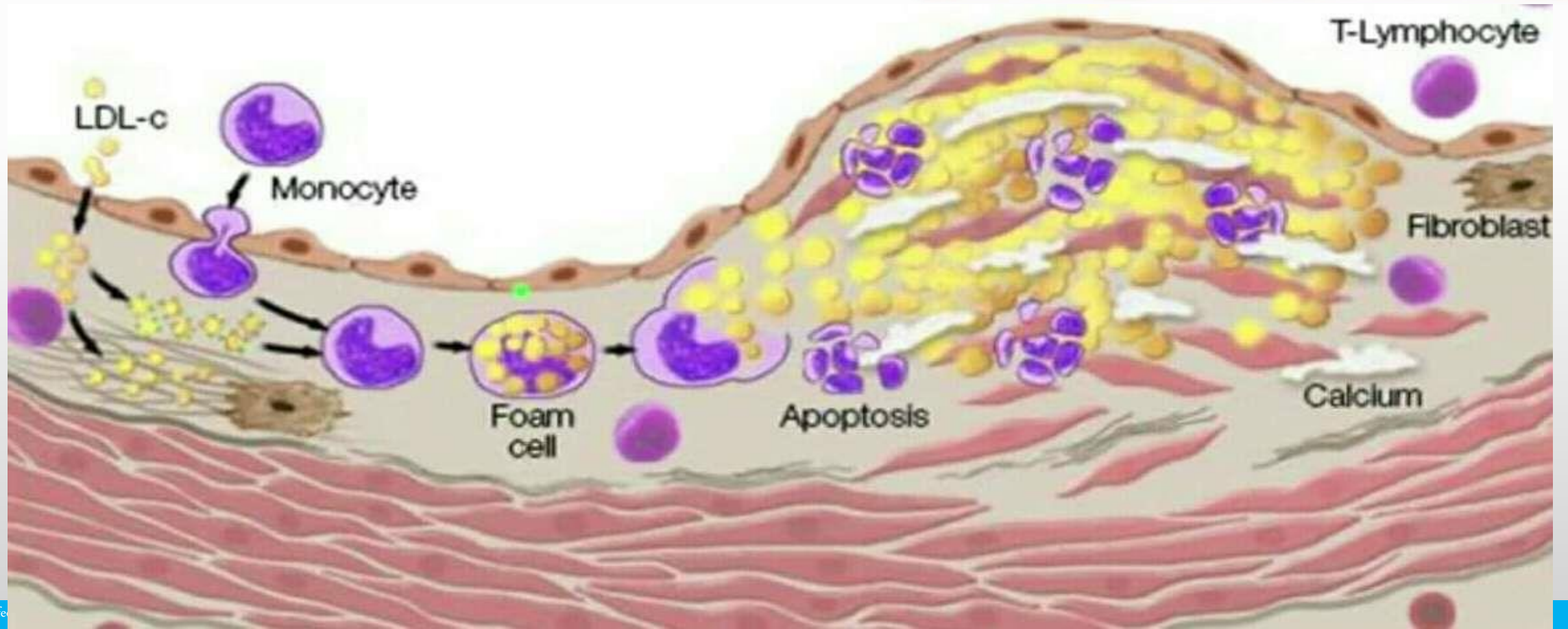
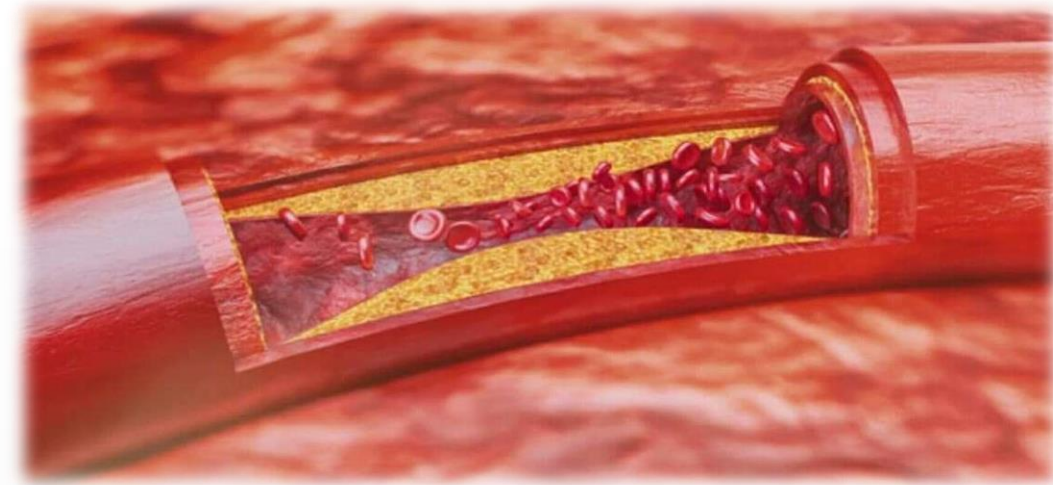
Foam cell



Foam cell



Atherosclerosis



Protein

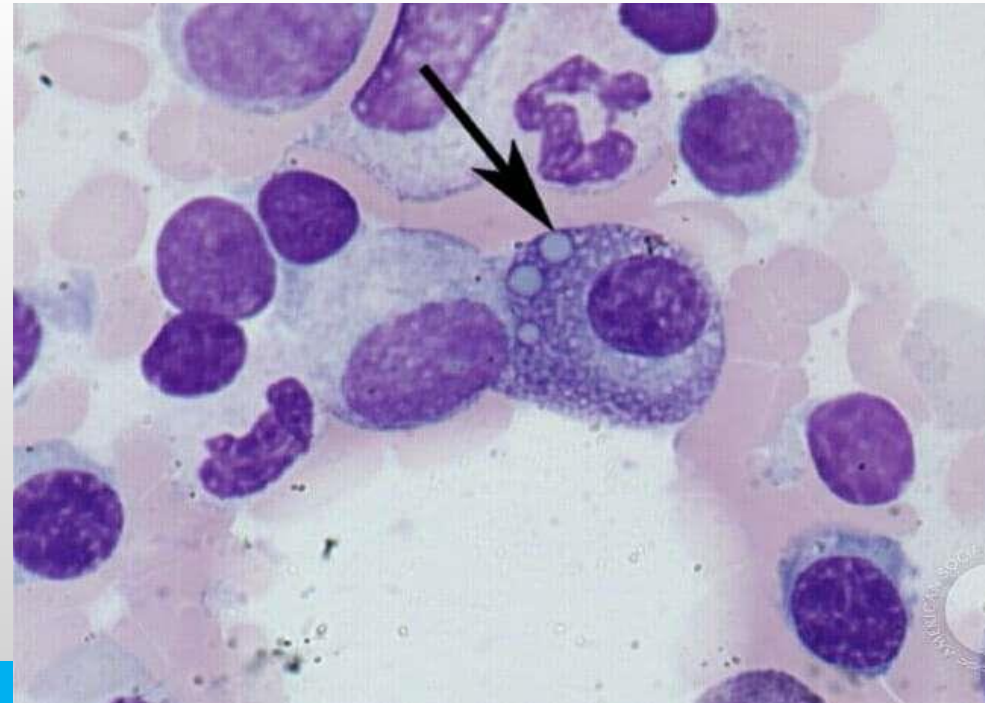
Protein accumulations may occur when excesses are presented to the cells or if the cells synthesize excessive amounts .

Examples :

In kidney disease like nephrotic syndrome, there is heavy protein leakage across the glomerular filter so there is a much larger reabsorption of the protein by pinocytosis in the proximal convoluted tubules, and vesicles containing this protein accumulate, giving the histologic appearance of pink, hyaline cytoplasmic droplets.

Another example is the marked accumulation of newly synthesized immunoglobulins that may occur in the rough endoplasmic reticulum (RER) of some plasma cells, forming rounded, eosinophilic Russell bodies .

Russell bodies



Glycogen

- Excessive intracellular deposits of glycogen are associated with abnormalities in the metabolism of either glucose or glycogen. In poorly controlled diabetes mellitus, glycogen accumulates in renal tubular epithelium, cardiac myocytes, and β cells of the islets of Langerhans.

Pigments

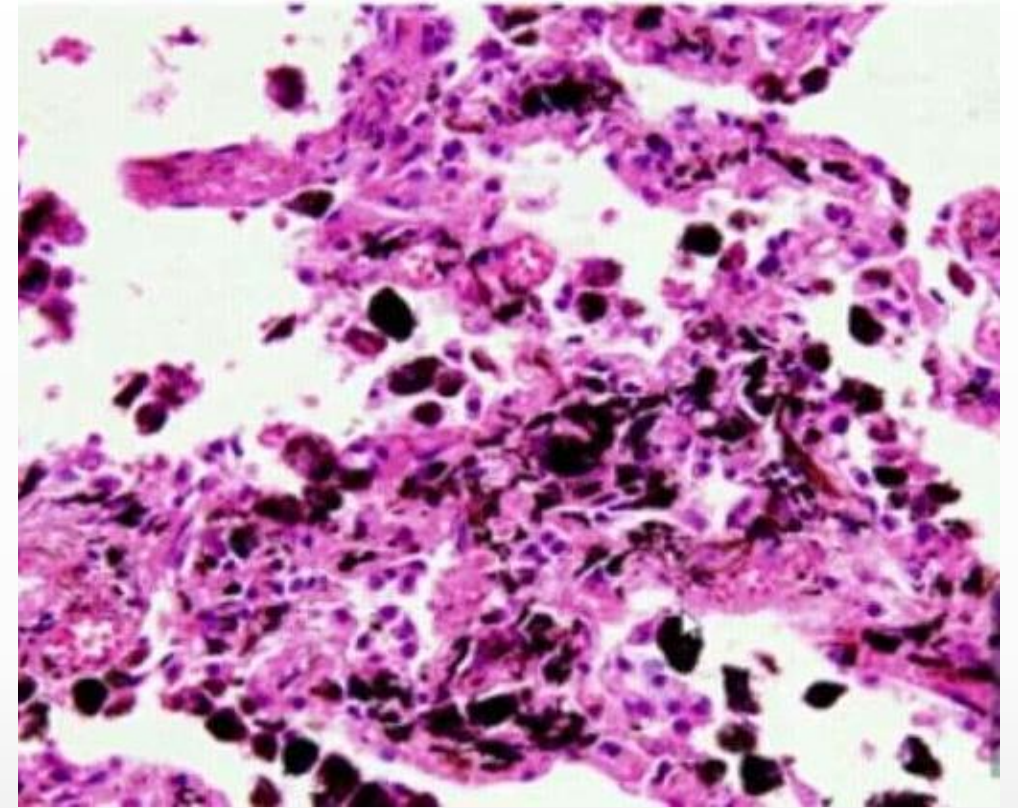
- Pigments are colored substances that are either
 - 1) **Exogenous**, coming from outside the body, such as carbon.
 - 2) **Endogenous**, synthesized within the body itself, such as lipofuscin, melanin, and certain derivatives of hemoglobin.

Exogenous pigment

- **Carbon** is the most common exogenous pigment is, a ubiquitous air pollutant of urban life. When inhaled, it is phagocytosed by alveolar macrophages giving black pigmentation of the lung. This condition is called **anthracosis**.



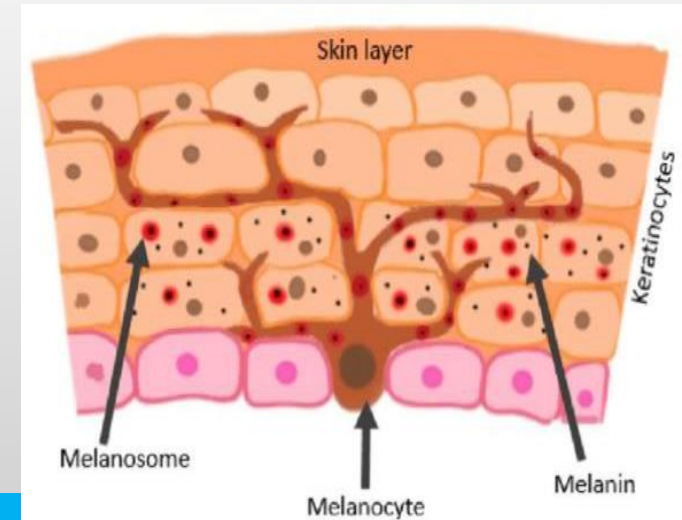
The black streaks seen between lobules of lung are due to accumulation of anthracotic pigment .



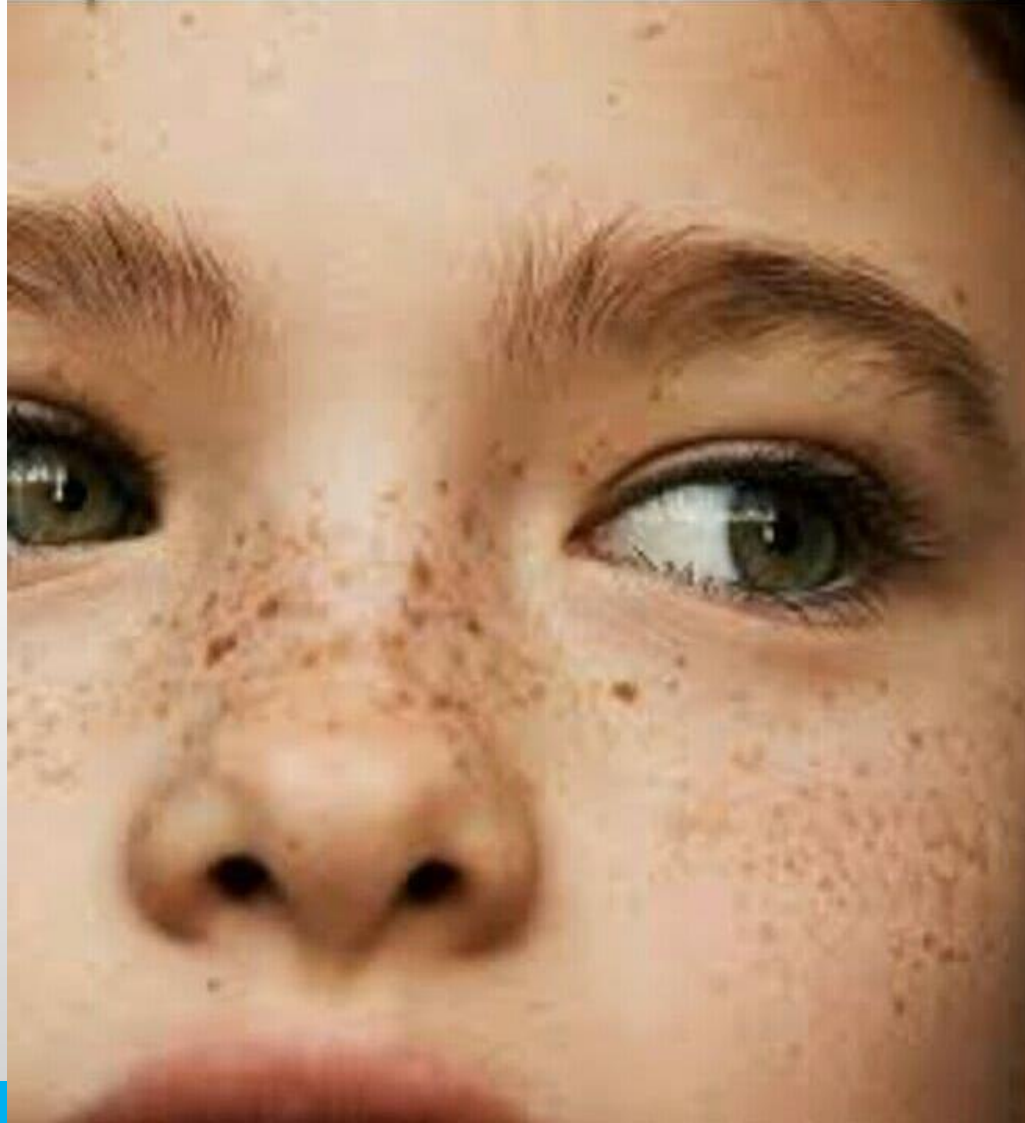
Here is anthracotic pigment in macrophages in a lung parenchyma

Endogenous pigment

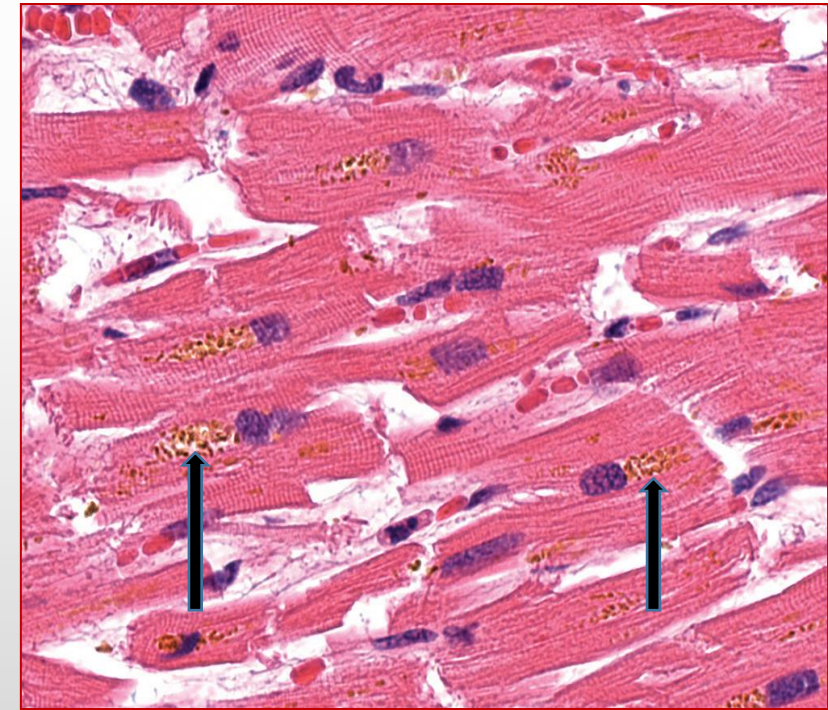
- **Melanin** is an endogenous, brown-black pigment that is synthesized by melanocytes located in the epidermis and acts as a screen against harmful UV radiation. Although melanocytes are the only source of melanin, adjacent basal keratinocytes in the skin can accumulate the pigment (e.g., in freckles), as can dermal macrophages .



Freckles



- **Lipofuscin**, or “wear-and-tear pigment,” is an insoluble brownish-yellow granular intracellular material that accumulates in a variety of tissues (particularly the heart, liver, and brain) with aging or atrophy. Lipofuscin represents complexes of lipid and protein .



Lipofuscin granules in cardiac myocytes
(deposits indicated by arrows.)

- **Hemosiderin** is a hemoglobin-derived granular pigment that is golden yellow to brown and accumulates in tissues when there is a local or systemic excess of iron. Although hemosiderin accumulation is usually pathologic, small amounts of this pigment are normal in the mononuclear phagocytes of the bone marrow, spleen, and liver, where aging red cells are normally degraded. Excessive deposition of hemosiderin, called **hemosiderosis**.



